Tutorial

A bank wants to predict loan default risk using logistic Regression.

Input features:

- credit score
- Annual income (in thousands)
- loan Amount (in thousands)
- Employment years.

Target variable: Default (Yes = 1, NO = 0)

complete dataset:

customer ID	credit Score	Annual Income	loan Amount	Employment Years	default
1	650	45	150	2	1
2	750	80	200	8	0
3	620	38	180	1	1
4	780	95	250	10	0
5	640	42	160	2	1
6	760	85	220	9	0
7	630	40	170	1	1
8	770	90	230	10	0
9	660	48	155	3	1
10	740	78	210	7	0

Sorted Nearest Neighbours:

Rank	customer	Distance	class
1	5	10.30	Y
2	1	30.56	Y
3	7	50.82	Y
4	3	61.12	Y
5	8	66.39	N

Classification of T_1

when $k=1$, Nearest Neighbour customer 5(Y)
predication = Y

when $k=3$, Nearest Neighbour customer
$(5,1,7) \rightarrow (Y,Y,Y)$
prediction = Y

when $k=5$, Nearest Neighbour customer
$(5,1,7,3,8) \rightarrow (Y,Y,Y,Y,N)$
Majority = Y
prediction = Y

Step1: Divide Dataset (k-fold cross validation)

Total = 20 sample

Default (Target variable) = 1 → [10 sample]

Default (Target variable) = 0 → [10 samples]

each fold = 4 samples ⇒ each fold consist of 2(1)s
2(0)s

Step(2):

fold 1 : customer 1(1), 2(0), 3(1), 4(0)

fold 2 : customer 5(1), 6(0), 7(1), 8(0)

fold 3 : customer 9(1), 10(0), 11(1), 12(0)

fold 4 : customer 13(1), 14(0), 15(1), 16(0)

fold 5 : customer. 17(1), 18(0), 19(1), 20(0)

$(5) = \sqrt{(20-22)^2 + (6-7)^2 + (3-4)^2 + (90-100)^2}$

$= \sqrt{4+1+1+100}$

$= 10.29$

$(6) = \sqrt{(20-6)^2 + (6-2)^2 + (3-0)^2 + (90-15)^2}$

$= \sqrt{(14)^2 + (4)^2 + (3)^2 + 5625}$

$= \sqrt{196 + 16 + 9 + 5625}$

$= 76.45$

$(7) = \sqrt{(20-28)^2 + (6-9)^2 + (3-6)^2 + (90-140)^2}$

$= \sqrt{64 + 9 + 9 + 2500}$

$= 50.81$

$(8) = \sqrt{(20-7)^2 + (6-3)^2 + (3-1)^2 + (90-25)^2}$

$= \sqrt{169 + 9 + 4 + 4225}$

$= 66.38$

Distance values

customer	values	Distance form	purchase
1	(25,8,5,120)	30.5	Y
2	(5,2,0,0)	91.37	N
3	(30,10,7,150)	61.09	Y
4	(8,3,1,20)	71.11	N
5	(22,7,4,100)	10.29	Y
6	(6,2,0,15)	76.45	N
7	(28,9,16,140)	50.81	Y
8	(7,3,1,25)	66.38	N

Step(3) Select fold (1)

Validation set (test) = customer (1, 2, 3, 4)

Training set = customer (5 to 20)

Step(4)

logistic Regression formula:

$$P = \frac{1}{1+e^{-z}}$$

$$z = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3 + \cdots + w_n x_n$$

$$z = \underset{\beta_0}{15} - \underset{\beta_1}{0.02}(\text{credit}) - \underset{\beta_2}{0.05}(\text{Income}) + \underset{\beta_3}{0.03}(\text{loan}) - \underset{\beta_4}{0.5}(\text{exmp})$$

calculate for each validation sample

customer 1

$$z = 15 - 0.02(650) - 0.05(45) + 0.03(150) - 0.5(2)$$

$$z = 3.25$$

$$P = \frac{1}{1+e^{-z}} = \frac{1}{1+e^{-3.25}} \cong 0.962$$

Step(7)

Calculation Matrix

$$\text{Accuracy} = \frac{\text{correct predictors}}{\text{Total predictors}} \times 100\% = \frac{4}{4} = 100\%$$

$$\text{precision} = \frac{TP}{TP+FN} = \frac{2}{2+0} = 1$$

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{2}{2+0} = 1$$

$$\text{F1-Score} = \frac{2 \times \text{precision} \times \text{Recall}}{\text{precision} \times \text{Recall}} = \frac{2 \times 1 \times 1}{1+1} = 1$$

Step(8)

Repeat these process for all folds [fold 1,2,3,4,5]

Folds	Accuracy
1	100%
2	100%
3	100%
4	100%
5	100%

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$$\text{Accuracy} = \frac{\text{Total correct}}{\text{Total}} \times 100$$

$$= \frac{2}{4} \times 100 = 50\%$$

Solution:

Step:1 Find Euclidean distance for

Test Instance $T_1 = (\overset{x_1}{20}, \overset{y_1}{6}, \overset{z_1}{3}, \overset{w_1}{90})$ Y

$T_2 = (10, 4, 2, 40)$ N $T_3 = (27, 9, 5, 130)$ Y

formula for Euclidean distance (d) $= \sqrt{(x_1-x_2)^2 + (y_1-y_2)^2 + (z_1-z_2)^2 + (w_1-w_2)^2}$

Distance

(1) $= \sqrt{(20-25)^2 + (6-8)^2 + (3-5)^2 + (90-120)^2}$

$= \sqrt{(25) + (4) + (4) + (900)}$

$= 30.5$

(2) $= \sqrt{(20-5)^2 + (6-2)^2 + (3-0)^2 + (90-0)^2}$

$= \sqrt{225 + 16 + 9 + 8100}$

$= 91.37$

(3) $= \sqrt{(20-30)^2 + (6-10)^2 + (3-7)^2 + (90-150)^2}$

$= \sqrt{100 + 16 + 16 + 3600}$

$= 61.09$

(4) $= \sqrt{(20-8)^2 + (6-3)^2 + (3-1)^2 + (90-120)^2}$

$= \sqrt{(12)^2 + (3)^2 + (2)^2 + (70)^2}$

$= \sqrt{144 + 9 + 4 + 4900}$

$= 71.11$

11	625	36	175	1	1
12	785	92	240	11	0
13	655	46	165	2	1
14	755	82	215	8	0
15	635	41	172	2	1
16	755	88	235	9	0
17	645	44	158	2	1
18	765	86	225	8	0
19	628	39	168	1	1
20	780	94	245	10	0

Questions:

(1) Divide the dataset into 5 folds (4 samples per fold) using stratified sampling. Show customer ID's and class distribution in each fold.

(2) For fold 1 as validation

- Train logistic regression on folds 2-5 (16 samples)
- logistic functions:

step(5)

Results	Actual	predicted	probability
customer1	1	1	0.962
customer 2	0	0	0.119
customer 3	1	1	0.996
customer 4	0	0	0.05

confusion matrix

	predicted 1	predicted 0
Actual 1	2 (TP)	0 (FP)
Actual 0	0 (FN)	2 (TN)

$$\text{precision} = \frac{TP}{TP+FP} = \frac{2}{2+0} = 100\%$$

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{2}{2+0} = 100\%$$

DT_1	DT_2	DT_3
52 A	153	37.42 A
113.3	11.57 R	195.89
103.15	203	22.93 A
126.98	39.87 R	205.58
22.9 A	118.8	72.31 A
90.62 R	11.4 A	173.02